Department of CSE
Jahangirnagar University
Savar, Dhaka-1342.

Course No: **Math 201** Class Test-**01** Time: **45** Minutes Marks: **30**

1. Evaluate $\oint_C \frac{e^{2z}}{(z-1)(z-3)^2}\,dz$; where C is the circle (a) $|z| = 2$, (b) $|z| = \frac{1}{2}$.

2. Expand $f(z) = \frac{1}{(z-1)(2-z)}$ in a Laurent series valid for $0 < |z-2| < 1$.

3. Evaluate $\int_0^\infty \frac{\sin 2x}{x}\,dx$.

Department of CSE
Jahangirnagar University
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Course No: Math 201 **Class Test-02** **Time: 30 Minutes** Marks: 30

1. Find the Fourier series expansion of $f(x) = e^{-x}; -\pi < x < \pi$.

2. Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$.

3. Show that $\underline{\nabla} r^n = n r^{n-2} \underline{r}$. Here, $\underline{r}$ is the position vector.

Department of Computer Science and Engineering
Jahangirnagar University
2nd Year 1st Semester Final Examination -2023
Course Code: MATH 201, Course Title: Mathematics III
Full Marks: 60, Time: 3 hours
[Answer any *six* questions]

1. a. If $\varphi(x,y,z) = 3x^2y - y^3z^2$, determine grad$\varphi$ at the point $(1,-2,-1)$. Find the directional derivatives of $\varphi = x^2yz + 4xz^2$ at the point $(1,-2,-1)$ in the direction $2\hat{\imath}-\hat{\jmath}-2\hat{k}$. 4

b. Determine *div grad* φ, where $\varphi = 2x^3y^2z^4$. Show that $\varphi = \frac{1}{r}$ is a solution of the Laplace's equation, $\underline{\nabla}^2\varphi = 0$, where $\underline{r}$ is the position vector. 4

c. Prove that curl grad $\varphi = \underline{0}$. 2

2. a. Show that $\underline{\nabla}r^n = nr^{n-2}\underline{r}$, where $\underline{r} = x\underline{i} + y\underline{j} + z\underline{k}$. 4

b. Define irrotational vector. Find the value of constants a, b, c so that 3

$$\underline{v} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$$

is irrotational.

c. If $\underline{v} = \underline{\omega} \times \underline{r}$, prove that $\underline{\omega} = \frac{1}{2}(\underline{\nabla} \times \underline{v})$ where $\underline{\omega}$ is a constant vector. 3

3. a. Discuss about the singularity of a complex function. Find the singularities of 3

$$f(z) = \frac{3z + 2}{(z-i)(z^2 + 16a^2)(z - 5i)^3(z + 1)^2}.$$

b. If $f(z)$ is analytic at every point on and within the simply closed curve C where a is within C then show that 4

$$\int_C \frac{f(z)}{z-a}dz = 2\pi i\, f(a).$$

c. Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z^2 - 9z + 20}dz$; where $C{:}\, z = 6e^{i\theta}; 0 \le \theta \le 2\pi$. 3

4. a. State and prove Laurent series. 5

b. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series valid for: 5

i. $|z| > 3$
ii. $0 < |z + 1| < 2$
iii. $|z| < 1$

5. a. Find the residues of $f(z) = e^z \operatorname{cosec}^2 z$ at all its poles in the finite plane. 3

b. Evaluate $\displaystyle\int_0^{\infty} \frac{\sin x}{x}\, dx.$ 5

c. State Taylor's theorem. 2

6. a. Find the Fourier series of the function, 5

$$f(x) = \begin{cases} 0; & -\pi < x < 0 \\ x; & 0 < x < \pi \end{cases}$$

b. Determine the Fourier coefficients of the following function 5

$$f(x) = \begin{cases} k; & -3 < x < 0 \\ -k; & 0 < x < 3 \end{cases}$$

7. a. Find the Fourier integral of the function $f(x) = e^{-kx}$ when $x > 0$ and $f(-x) = f(x)$ for $k > 0$, and hence prove that $\int_0^{\infty} \frac{\cos ux}{k^2+u^2} du = \frac{\pi}{2k} e^{-kx}$. 6

b. Determine the finite Fourier cosine transform of the function 4

$$F(x) = \begin{cases} 1, & 0 < x < \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x < \pi \end{cases}$$

8. a. Define Laplace transform and Inverse Laplace transform of a function. Find the Laplace transform of the function $F(t) = te^{-t} \sin 2t$. 4

b. Solve the following differential equations by using Laplace transform 6

i. $y' + 3y = t^2 e^{-3t} + te^{-2t} + t, \quad y(0) = 1$

ii. $y'' - 3y' + 2y = 4, \quad y(0) = 2, y'(0) = 3.$

CSE 203 Computer Ethics and Cyber Law – Tutorial 1 Mark 20

Consider the following scenario:

> *"Farid is a second-year CSE student at Narsingdi University. He depends on private tuition-based income to bear his daily life and academic expenses. Recently his father passed away and now he has to support his family too. He finds it hard to find time for study and his grades are getting low. One day, while discussing with a faculty, he could oversee his password. Later he logged in to the account and could access the course materials including old and current tutorial and final examination question scripts. He prepared himself with it and achieved A+ in the course taken by the Faculty."*

1. According to "Cyber Security Act-2023", analyze the above scenario and (i) identify the rules that have been broken (if any) by Mr. Farid and (ii) the corresponding punishments he will be subject to. *[7 marks]*
2. Evaluate the act of Mr. Farid using (i) Virtue Ethics and (ii) Act Utilitarianism ethical theories. *[7 marks]*
3. Differentiate among morals, ethics, and law. Point out few concerns about Ethical Use of Information Technology. *[6 marks]*

CSE 203 Computer Ethics and Cyber Law – Tutorial 1 Mark 20 (5x4)

Q1. Explain the ethical relationship between IT professionals and Suppliers. Distinguish between a bribe and a gift.

Q2. Use the five-step decision-making process discussed to analyze the following situations and recommend a course of action:

"It appears that someone is using your firm's corporate directory—which includes job titles and email addresses—to contact senior managers and directors via email. The email requests that the recipient click on a URL, which leads to a Web site that looks as if it were designed by your Human Resources organization. Once at this phony Web site, the employees are asked to confirm the bank and account number to be used for electronic deposit of their annual bonus check. You are a member of IT security for the firm. What can you do?"

Q3. To ensure the security of their customers and products, many business stores have started using body scanners. How can information collected by body scanners installed in stores affect customer privacy?

Q4. Should You Copyright or Patent Software? Explain in detail.

CSE 203 Computer Ethics and Cyber Law – Tutorial 3 Mark 20

Q1. Identify the attributes of a mature profession. Prepare an unjust scenario in your society/ community/ organization (not necessarily true) and answer your decision to be a whistle-blower or not according to Richard DeGeorge's questions for whistle-blowing. [2 +4 marks]

Q3. You are a 4th Year 2nd semester student. You sent your resume to a half-dozen companies hoping to get a job. A month ago you interviewed at ABC Corporation and XYZ Corporation. Two weeks ago XYZ Corporation offered you a job. One week ago you accepted their offer, agreeing to start work a month after graduation. Today you received a much better offer from ABC Corporation. What should you do (review/analyze the software engineering code of ethics)? [8 marks]

Q4. Write short notes on the following: (i) Green computing (ii) Digital Divide (iii) Impact of globalization on the IT industry. [6 marks]

Jahangirnagar University

Department of Computer Science and Engineering

2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2023

Course Title: Computer Ethics and Cyber Law
Time: 3 Hours

Course No: **CSE-203**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1. Answer ***All of the following***.

   a) **Define** cyber law and cyber space. 2
   b) **What** is corporate social responsibility (CSR), and **why** is fostering good business ethics important? 2
   c) **Describe** a hypothetical situation in which the action you would take is not legal, but it is ethical. Also, describe a hypothetical situation where the action you would take is legal, but not ethical. 2
   d) **Define** the following security terms: (i) Worm (ii) Backdoor Trojan (iii) DoS attack (iv) Cyberterrorists. 2
   e) **Define** the following terms related to intellectual property: (i) Fair use (ii) Creative common 2
   f) **State** the Authorship and Assignment issues in cyber law. 2

2. Answer ***Any Four*** out of ***Five***.

   a) **Illustrate** the measure that we use in fight back against cyberbullying as the recent report on platform like Facebook and Twitter. 3
   b) Should software piracy within the boundaries of third-world countries be tolerated to allow these countries an opportunity to move more quickly into the information age? **Why** or why not? 3
   c) **Explain** the ethical relationship between IT professionals and Suppliers. Distinguish between a bribe and a gift. 3
   d) Answer the followings: 3
   (i) Is it morally acceptable to use a denial-of-service attack to shut down a Web server that distributes child pornography? [1 marks]
   (ii) Some IT security personnel believe that their organizations should employ former computer criminals to identify weaknesses in their organizations' security defenses. Do you agree? Why or why not? [2 marks]
   e) Answer the followings: 3
   (i) What is spyware? **Give** three examples of how data mining is being used on information collected from social networks. [2 marks]
   (ii) Should You Copyright or Patent Software? **Explain**. [1 mark]

3. Answer ***Any Three*** out of ***Four***.

   a) **Differentiate** among morals, ethics, and law. Discuss some situations in which the benefits of breaking the law are greater than the harms. 4
   b) **Describe** similarities and differences between Kantianism and rule utilitarianism. 4
   c) **Compare** between: (i) Proprietary and Open Source software (ii) hacker and cracker. 4

d) Construct a comparison table for active cyber-attack and passive cyber-attack. Explain the differences between black hat hackers and white hat hackers. 4

4. Answer *Any Three* out of *Four*.

a) *"You are a recent graduate of a well-respected business school, but you are having trouble getting a job. You worked with a professional résumé service to develop a well-written résumé and placed it on several Web sites; you also sent it directly to contacts at a dozen companies. So far, you have not even had an invitation for an interview. You know that one of your shortcomings is that you have no real job experience to speak of. You are considering beefing up your résumé by exaggerating the extent of the class project you worked on for a few weeks at your brother-in-law's small consulting firm. You could reword the résumé to make it sound as if you were actually employed and that your responsibilities were greater than they actually were. What would you do?"* 4

Use the five-step decision-making process to **analyze** the above situation and **recommend** a course of action. Which option would you pursue and why? And why do you not **choose** the other options?

b) *"You are a new brand manager for a product line of Coach purses. You are considering purchasing customer data from a company that sells a large variety of women's products online. In addition to providing a list of names, mailing addresses, and email addresses, the data includes an estimate of customers' annual income based on the zip code in which they live, census data, and highest level of education achieved. You could use the data to identify likely purchasers of your high-end purses, and you could then send those people emails announcing the new product line and touting its many features."* 4

i. List the advantages and disadvantages of such a marketing strategy. [1 mark]
ii. Would you recommend this means of promotion in this instance? Why or why not? Use the *five-step decision-making process* to **analyze** the above situation and **recommend** a course of action. [3 marks]

c) *"You are a 4th Year 2nd semester student. You sent your resume to a half-dozen companies hoping to get a job. A month ago you interviewed at ABC Corporation and XYZ Corporation. Two weeks ago XYZ Corporation offered you a job. One week ago you accepted their offer, agreeing to start work a month after graduation. Today you received a much better offer from ABC Corporation."* 4

What should Gian do? Consider the *software code of ethics* to **analyze** the situation and **recommend** a course of action.

d) Google has come out with a revolutionary new product known as Google Glass. Although it is highly-priced, it has very interesting features that have made people take note of it. It has generated a lot of excitement and has made its way into the wish-list of must-have gadgets of many people. Explore the Internet and find out the different features of Google Glass. 4

(i) What are cookies, and why are they used?

(ii) What makes it different from other similar products as Google Glass?

(iii) In what ways may Google Glass prove to be a dangerous tool?

(iv) Is it possible that this new device may be a cause of concern when it comes to privacy issues?

5. Question No. 5 will be based on ***CO5***. This question will be consisting of ***Three*** Sections. Each section may contain several subsections like (a (i), a(ii), a(iii) etc)
Students have to answer ***Any Two*** out of ***Three***.

a) Consider a cybercrime scenario as below: 6

"A worm is a self-contained program that spreads through a computer network by taking advantage of security holes in the computers connected to the network. Recently, an worm named ABC created by Mr. Ruhul infected many computers running the Windows 8, and Windows 10 operating systems. The worm caused computers it infected to reboot every few minutes.
Soon another worm named XYZ created my Mr. Kamal was exploiting the same security hole in Windows to spread through the Internet. However, the purpose of the new worm, was benevolent. Since it took advantage of the same security hole as ABC, it could not infect computers that were immune to the ABC worm. Once XYZ gained access to a computer with the security hole, it located and destroyed copies of the ABC worm. It also automatically downloaded from Microsoft a patch to the operating system software that would fix the security problem. Finally, it used the computer as a launching pad to seek out other Windows PCs with the security hole."

Considering the above scenario, answer the followings:

(i) Was the action of the person who released the XYZ worm morally right or wrong? [1 mark]

(ii) **Judge** the action of Mr. Kamal (i) using "Act Utilitarianism" ethical theory. [2 marks]

(iii) Consider the above scenario is staged in the context of Bangladesh. Now, according to *Cyber Security Law, 2023,* analyze the above scenario and identify the rules that have been broken by Mr. Ruhul (if any) and the corresponding punishments he will be subject to. [3 marks]

b) *"A person's identity is their most valuable possession, especially in the age of online shopping and banking. Consumers use their address, phone number and other personal details to keep transactions secure. That's why identity theft – when this information gets stolen – can have an enormous impact.* 6
In the US, the largest ever case of identity theft occurred when an employee at a software company sold consumer data to a network of criminals. The passwords and codes enabled the gang to download thousands of credit reports – and to cause losses of between US$ 50 to 100 million by using the information to perform financial transactions. But identity theft isn't always an inside job: In 2009, hackers took advantage of security weak spots to steal 45.5 million debit card numbers from a payment processing company. Within just twelve hours, they had withdrawn more than US$9.4 million from around 2,100 cash machines in 280 countries worldwide."

Answer the followings:

i) Following the above story, **create** a real-like cybercrime scenario in the context of Bangladesh relating identity theft. [2 marks]

ii) **Judge** the scenario in (i) considering one or more appropriate ethical theory. [2 marks]

iii) Do you think the current *Cyber Security Law* in Bangladesh will cover the cybercrime in (i)? If so, **identify** the clauses are in consideration for this crime and the corresponding offences and punishments. [2 marks]

9) Answer the followings: 6

(i) Consider the software code of ethics to **analyze** the following situation and **recommend** a course of action. [4 marks]

(ii) Also **judge** the situation using Richard DeGeorge's questions for whistle-blowing. [2 marks]

"Company X wants to open a dating service Web site. It hires Company Y to develop the software. Company Y hires Gina as a private contractor to provide a piece of instant messaging software for the package. Gina's contract says she is not responsible for the security of the site. Company Y is supposed to perform that bit of programming. However, software development runs behind schedule, and Company Y implements a simplistic security scheme that allows all messages to be sent in plain text, which is clearly insecure.

Gina brings her concerns to the management of Company Y. Company Y thanks her for her concern, but indicates it still plans to deliver the software without telling Company X. Company Y reminds Gina that she has signed a confidentiality agreement that forbids her from talking about the software to anyone, including Company X."

------ End of question script -----

Jahangirnagar University

Department of Computer Science and Engineering

Second Year First Semester B. Sc.(Hons) Tutorial Examination-01

Course Code: CSE-205 Title: Numerical Methods

Time allowed: 45 minutes Total Marks: 20

1. Derive the Newton-Raphson formula using Taylor series expansion. Estimate the root of the equation 2x2-3x+5=0 using Newton-Raphson method (show at least five iterations). [10]

2. Differentiate between Newton-Raphson method and Secant method. [4]

3. Explain the principle of bisection method with the help of an illustration. [6]

Jahangirnagar University
Department of Computer Science and Engineering
Second Year First Semester B. Sc.(Hons) Tutorial Examination-02

Course Code: CSE-205 | Title: Numerical Methods

Time allowed: 45 minutes | Total Marks: 20

1. Differentiate between Jacobi iteration method and Gauss-Seidel method. [5]
2. Solve the following system of equations by Gauss-Seidel iteration method (show at least 4 iterations): [7.5]

$$x_1 + 2x_2 - 3x_3 = 4$$
$$2x_1 + 4x_2 - 6x_3 = 8$$
$$x_1 - 2x_2 + 5x_3 = 4$$

3. Solve the following system of equations using partial pivoting technique: [7.5]

$$2x_1 + 2x_2 + x_3 = 6$$
$$4x_1 + 2x_2 + 3x_3 = 4$$
$$x_1 - x_2 + x_3 = 0$$

CT-03

Course Code: CSE 205, Course Title: Numerical Methods

2^{nd} Year 1^{st} Semester, Dept. of CSE

Jahangirnagar University

<u>Questions</u> Marks: 20

1. What is regression? Why is it necessary?
2. What is the principle of least squares regression? Apply the least square regression to fit a straight line to the following set of data.

x:	1	3	4	6	8	9	11
y:	1	2	4	4	5	7	8

3. Given below is a table of data for log x. Estimate log 2.5 using Lagrange interpolation polynomial.

i	0	1	2	3
x_i	1	2	3	4
$\log x_i$	0	0.3010	0.4771	0.6021

x_0 x_1 x_2 x_3

4. What is the major pitfall of using Lagrange polynomial?

CT-04

Course Title: Numerical Methods

Course Code: CSE 205

Total Marks: 15

1. Why do we need to use numerical techniques to obtain the estimates of function derivatives?

2. Use two Segment Trapezoidal Rule to find the area under the curve

$$f(x) = \frac{300x}{1+e^{x}} \quad \textit{from} \quad x = 0 \ \textit{to} \ x = 10$$

3. Estimate approximate derivative of f(x) = x2 at x=1, for h = 0.2, 0.1, 0.05, 0.01 using the first-order forward difference formula.

Jahangirnagar University
Department of Computer Science and Engineering
2^{nd} Year 1^{st} Semester B.Sc. (Hons.) Final Examination -2023

Course Title: Numerical Methods
Time: 3:00 Hours

Course No: CSE-205
Full Marks: 60

Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.

1.

Section-I (CO1)
Answer ***All of them.***

a) ***List out*** the limitations of using direct methods for solving a system of linear equations. 2
b) ***Define***: i) Regression ii) Interpolation 2
c) ***List out*** the methods for solving non-linear equations with appropriate examples. 2
d) **Define** numerical integration. Why is it used instead of analytical methods for integration? 2
e) ***State*** Simpson's 1/3 rule (Three-point formula). 2
f) ***State*** the major pitfall of using Lagrange polynomial. 2

2.

Section-II (CO2)
Answer ***Any Three*** out of ***Four.***

a) ***Explain*** the principle of Secant method with the help of an illustration. 4
b) ***Illustrate*** Taylor method to solve the following differential equation: 4

$$y' = x^2 + y^2$$

Where, $x = 0.25$ and $x = 0.5$, given y (0) =1.

c) ***Differentiate*** between Jacobi iteration method and Gauss-Seidel method. 4
d) **Apply** the least square regression to the following set of data to fit a straight line: 4

x:	1	3	4	6	8	9	11
y:	1	2	4	4	5	7	8

3.

Section-III (CO3)
Answer ***Any Three*** out of ***Four.***

a) **Apply** shooting method to solve the following equation: 4

$$\frac{d^2y}{dx^2} = 6x, \qquad y(1) = 2 \quad and \quad y(2) = 9$$

in the interval (1, 2).

b) **Use** the Secant method to *solve* the root of the equation: $f(x) = x^2 - 4x - 10$ with the initial estimates of $x_1 = 4$ and x_2=2. ***Show*** at least four iterations. 4
c) **Compute** approximate derivative of f(x) = x^2 at x=1, for h = 0.2, 0.1, 0.05, 0.01 using the first-order forward difference formula. 4

d) ***Demonstrate*** Newton's divided difference interpolation polynomial. Also **show** some merits and demerits of this method. 4

4.

Section-IV (CO4)

Answer ***Any Three*** out of ***Four.***

a) ***Solve*** the following system of equations by Gauss-Seidel iteration method (show at least five iterations): 4

$$x_1 + 2x_2 - 3x_3 = 4$$
$$2x_1 + 4x_2 - 6x_3 = 8$$
$$x_1 - 2x_2 + 5x_3 = 4$$

b) ***Classify*** the four possible solution conditions of a system of linear equations. Analyze each one of them with proper illustration. 4

c) Illustrate the false position method to find a root of the equation: $x^2-x-2=0$ in the range $1<x<3$. 4

d) Apply the Lagrange interpolation polynomial to fit the followings data: 4

i	0	1	2	3
x_i	0	1	2	3
e^x-1	0	1.7183	6.3891	19.0855

Use the polynomial to estimate the value of $e^{1.5}$

5.

Section-V (CO5)

Answer ***Any Two*** out of ***Three.***

a) For the following differential equation, **estimate** $y(2)$ using i) Euler's method and ii) Heun's method using $h = 0.25$. Compare the results with exact answers. 6

$$y'(x) = 2y/x \qquad \text{with } y(1) = 2.$$

b) The values of distance travelled by a car at various time intervals are given in the following table. ***Estimate*** the velocity at time $t = 5$, $t = 7$ and $t = 9$. 6

Time, t(s)	5	6	7	8	9
Distance travelled, s(t) (km)	10.0	14.5	19.5	25.5	32.0

c) ***Estimate*** the integral $\int_{-1}^{1} e^x\,dx$ using composite trapezoidal rule for (i) n = 2 and (ii) n = 4 with an accuracy to five decimal places. 6

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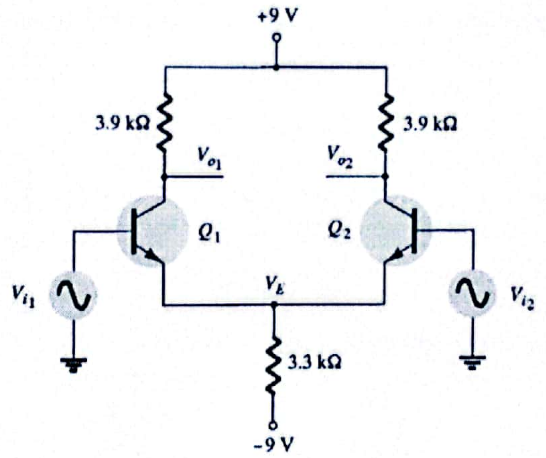
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. Final Examination -2023

Tutorial 1
Course Title: **Electronic Devices and Circuits II**
Time: **60 minutes**

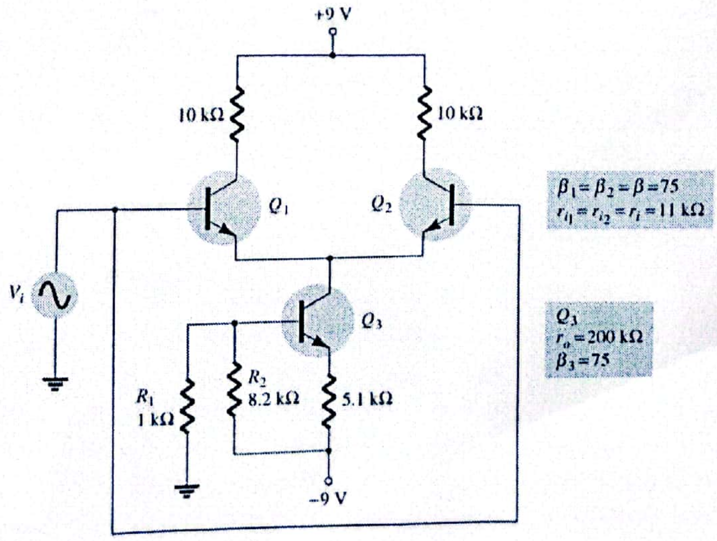
Course No: **CSE-207**
Full Marks: **30**

[Answer **ALL** of the following questions]

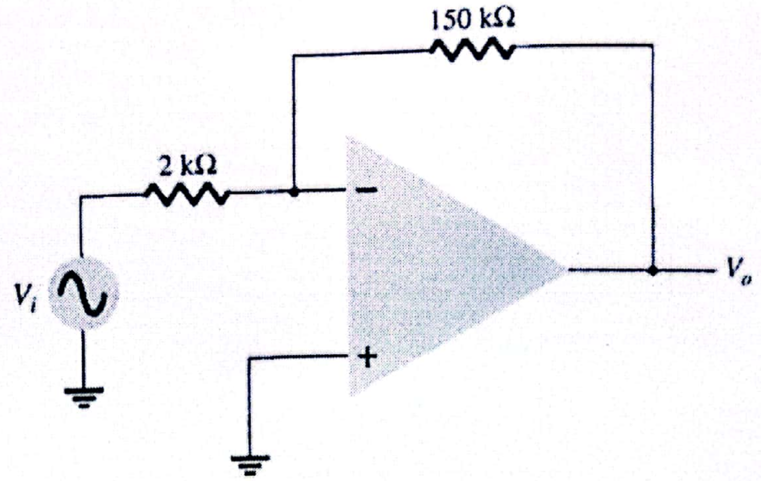
1. Explain with necessary diagrams and equations on *Op-amp Basics (Any TWO).* 5
   (i) Op-amp (ii) Virtual ground (iii) Common mode rejection
2. Explain with necessary diagrams and equations on *Practical Op-amp Circuits* (*Any TWO*). 5
   (i) Unity follower (ii) Summing amplifier (iii) Integrator
3. Explain with necessary diagram and equations on Op-amp Frequency Parameters (*Any TWO*). 5
   (i) Gain-Bandwidth (ii) Slew Rate (iii) Maximum Signal Frequency
4. Calculate the dc voltages and currents in the circuit. 5



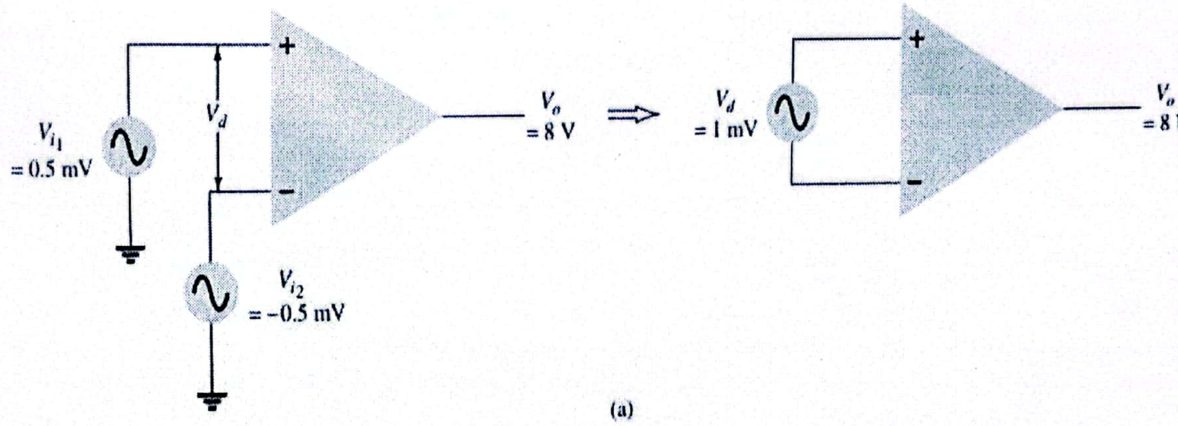
5. Calculate the common-mode gain for the differential amplifier. 2.5



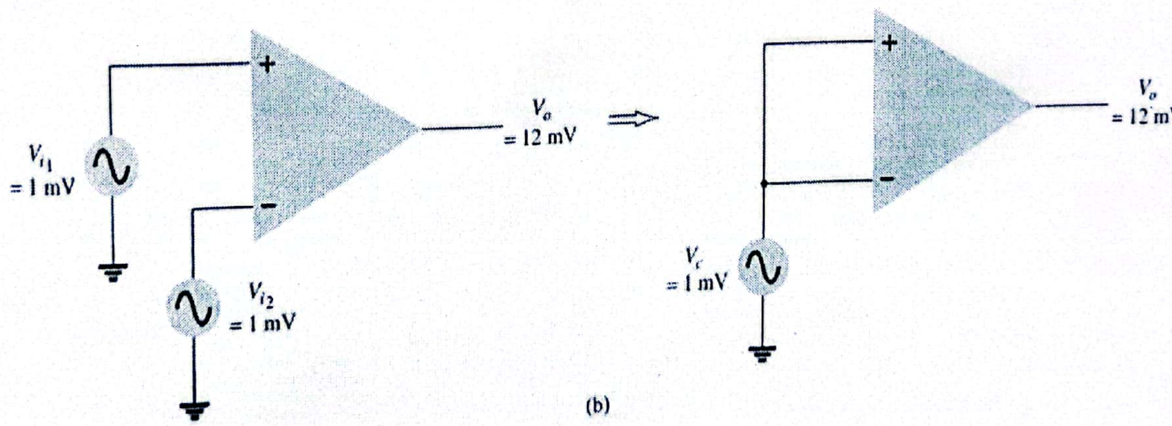
6. Calculate the output offset voltage of the circuit. The op-amp spec lists $V_{IO} = 1.2\ mV$. 2.5



7. Calculate the CMRR for the circuit measurements. 5



(a)



(b)

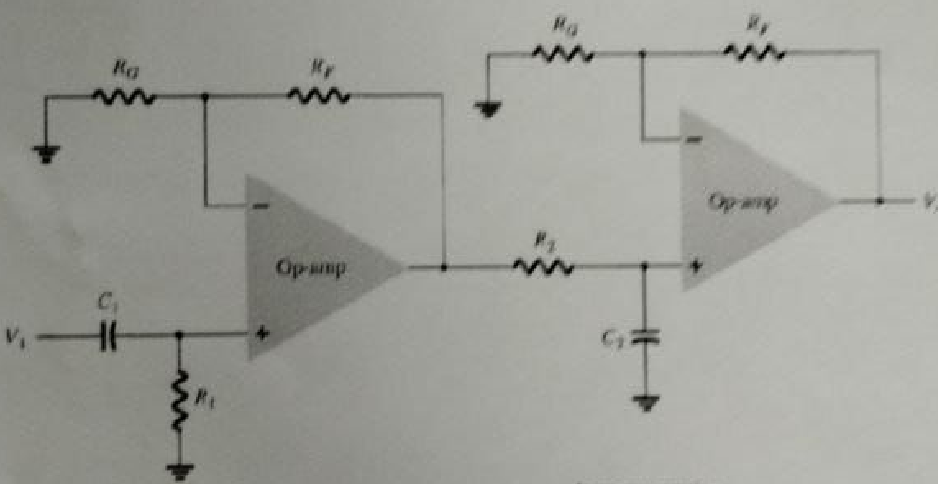
Jahangirnagar University
Department of Computer Science and Engineering
2^{nd} Year 1^{st} Semester B.Sc. Final Examination -2023

Tutorial 2
Course Title: **Electronic Devices and Circuits II**
Time: **30 minutes**

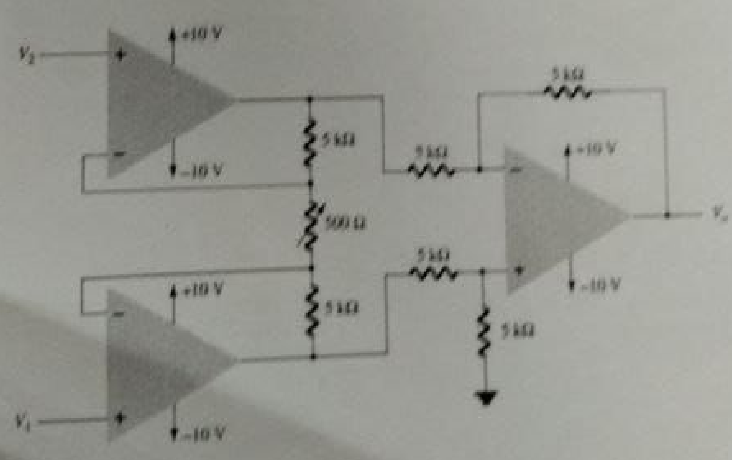
Course No: CSE-207
Full Marks: 30

[Answer **ALL** of the following questions **SEQUENTIALLY**]

1. Write short notes on with circuit diagram and equations: 7.5
   (i) Multi-stage Gain
   (ii) Voltage Summing
   (iii) Voltage Buffer
2. Derive the voltage output equation of an instrumentation amplifier. 5
3. Explain low-pass, high-pass and band-pass active filters with circuit diagrams, equations and response plot. 7.5
4. Calculate the cutoff frequencies of the bandpass filter circuit of the following figure with $R_1 = R_2 = 15K\Omega$, $C_1 = 0.15\ \mu F$ and $C_2 = 0.002\ \mu F$. 5



5. Calculate the voltage V_1, V_2 for the following circuit where the output voltage is 56.7V and $V_1 = 2V_2$. 5



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Examination 2023

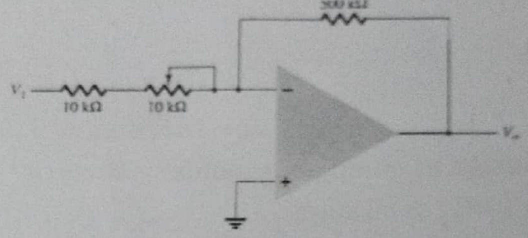
Course Title: **Electronic Devices and Circuits-II** Course No: **CSE 207**
Time: **3 hours** Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1. Question No. 1 will be based on ***CO1***.

Answer ***All of them***.

a) What is the range of the voltage-gain adjustment in the circuit of Fig. 10.63? 2



b) Explain the maximum signal frequency at which an op-amp may operate. 2

c) Write a short note on the use of buffer amplifiers to provide output signals. 2

d) Define the harmonic distortion of a power amplifier. 2

e) Explain the phase-shift oscillator with a circuit diagram. 2

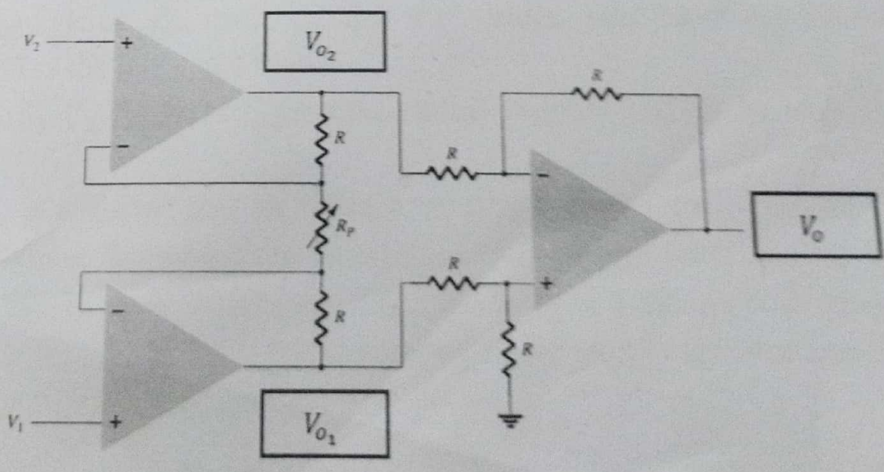
f) What is the ripple factor of a sinusoidal signal having peak ripple of 2 V on an average of 50 V? 2

Question No. 2 will be based on ***CO2***.

Answer **Any Three** out of **Four**.

a) Derive the equation $\frac{V_0}{V_1} = -\frac{R_f}{R_1}$ for an op-amp circuit from the concept of virtual ground. 4

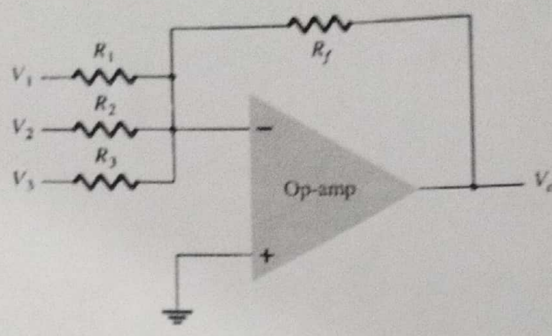
b) Explain the instrumentation amplifier. Derive the equation of the output voltage V_o for an instrumentation amplifier by mentioning the intermediate equations for V_{o_1}, and V_{o_2}. 4



4

c) Differentiate between classes A, B, AB, C, and D amplifiers. 4

d)



Calculate the output voltage developed by the circuit of this Figure for $V_1 = +0.2V$; $V_2 = -0.5$ V; $V_3 = +0.8V$; $R_1 = 33K$; $R_2 = 22k$; $R_3 = 12K$ and $R_f = 330$ kΩ.

3.

Question No. 3 will be based on ***CO3***.

Answer ***Any Three*** out of ***Four***.

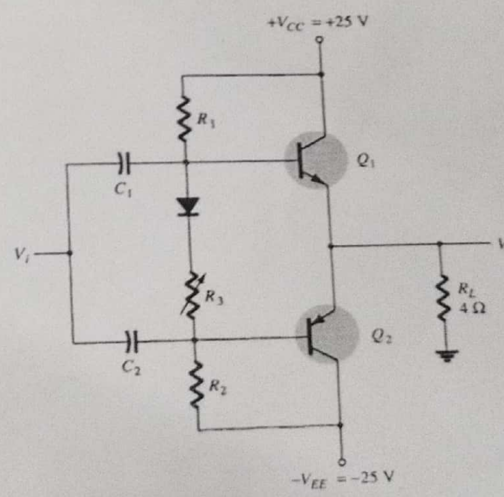
a) Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use $R_f = 1$ MΩ in all cases. 4

(i) $V_1 = +1$ V, $V_2 = +2$ V, $V_3 = +3$ V, $R_1 = 500$ kΩ, $R_2 = 1$ MΩ, $R_3 = 1$ MΩ

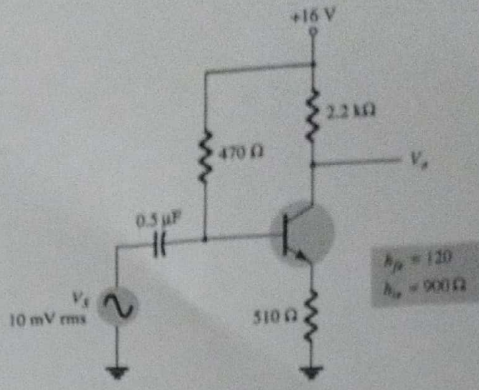
(ii) $V_1 = -2$ V, $V_2 = +3$ V, $V_3 = +1$ V, $R_1 = 200$ kΩ, $R_2 = 500$ kΩ, $R_3 = 1$ MΩ.

b) Show the annotated circuit diagram of three op-amp stages using an LM348 IC to provide outputs that are 10, 20, and 50 times larger than the input. Use a feedback resistor $R_f = 500\ k\Omega$ in all stages. 4

c) For the circuit, calculate the input power, output power, and power handled by each output transistor and the circuit efficiency for an input of 12 V rms. 4

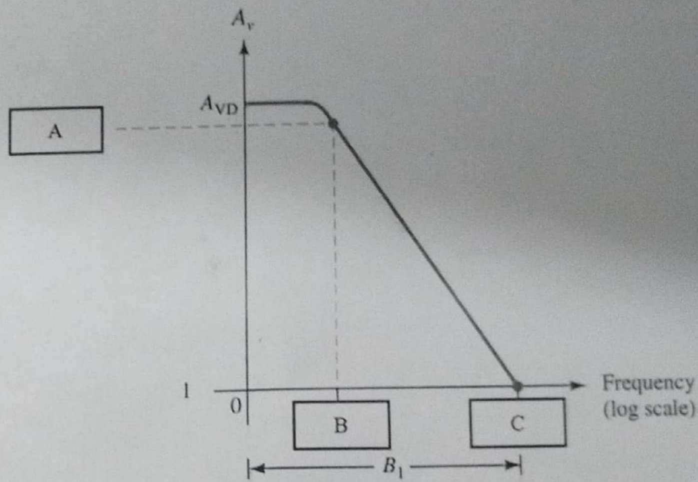


d) Calculate the voltage gain of the BJT amplifier circuit with current-series feedback.



Question No. 4 will be based on ***CO4***.
Answer ***Any Three*** out of ***Four***.

4. a) Explain the plot by filling up the missing op-amp parameters in square boxes A, B, and C. Establish the relationship of parameters in A, B, and C. 4



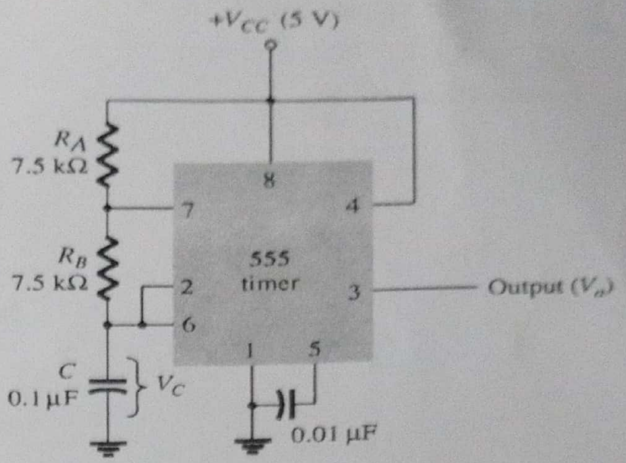
b) Design a bandpass filter where the ratio of cut-off frequencies is $\frac{f_{OH}}{f_{OL}} = \frac{37}{3}$. You can consider any values $R_1, R_2, C_1, and\ C_2$ for constructing the bandpass filter that satisfies the ratio. 4

(i) Calculate the cut-off frequencies $f_{OL}\ and\ f_{OH}$ for the values you have considered $R_1, R_2, C_1, and\ C_2$ that satisfies the mentioned ratio.

(ii) Show the values of $R_1, R_2, C_1, C_2, f_{OL}, and\ f_{OH}$ on the circuit diagram and on the response plot of the bandpass filter.

c) A voltage-controlled oscillator (VCO) is a circuit that provides a varying output signal (typically of square-wave or triangular-wave form) whose frequency can be adjusted over a range controlled by a dc voltage. An example of a VCO is the 566 IC unit, which con- tains circuitry to generate both square-wave and triangular-wave signals whose frequency is set by an external resistor and capacitor and then varied by an applied dc voltage. Explain operation of any VCO IC. 4

d) Determine the frequency and draw the output waveform for the stable multivibrator circuit using the 555 timer IC. 4



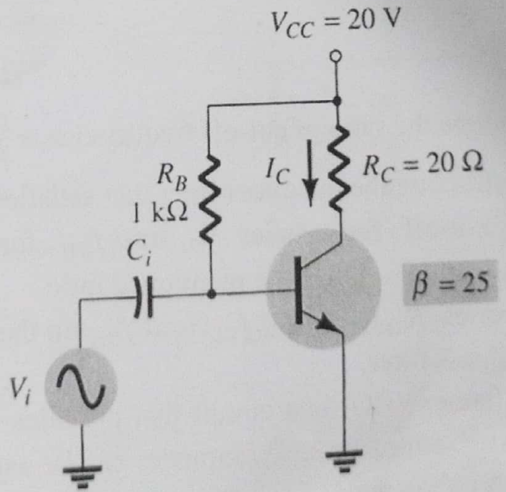
Question No. 5 will be based on ***CO5***.

Answer ***Any Two*** out of ***Three***.

5. a) "The larger the common-mode rejection ratio (CMRR), the closer is the output voltage (V_o) to the input voltage difference (V_d) times the difference gain. (A_d)" – Evaluate the statement theoretically with mathematical notations. Justify your answer by determine the output voltage of an op-amp for input voltages of $V_{i_1} = 150\ \mu V$ and $V_{i_2} = 140\ \mu V$. Consider the amplifier has a differential gain of $A_d =$ 4000 and the value of CMRR is 6

(i) 100
(ii) 1000
(iii) 10000

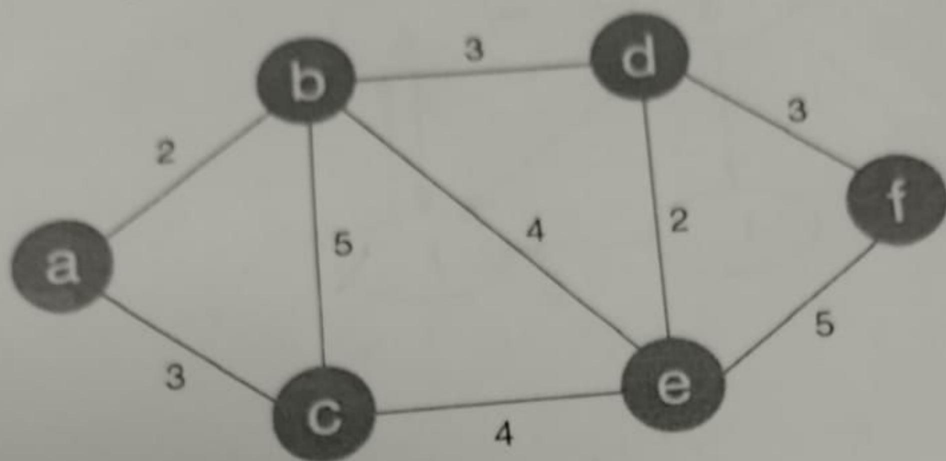
b) Prove that the maximum efficiency of a class A series-fed amplifier is 25%. Calculate the input power, output power, and efficiency of a Class A series-fed amplifier for an input voltage that results in a base current of 10 mA peak. 6



c) Explain the construction of the Wien bridge oscillator circuit using op-amp amplifiers with necessary equations. Design the RC elements of a Wien bridge oscillator for operation at resonance frequency and annotate the oscillator circuit with RC element values. 4

Name: Roll:

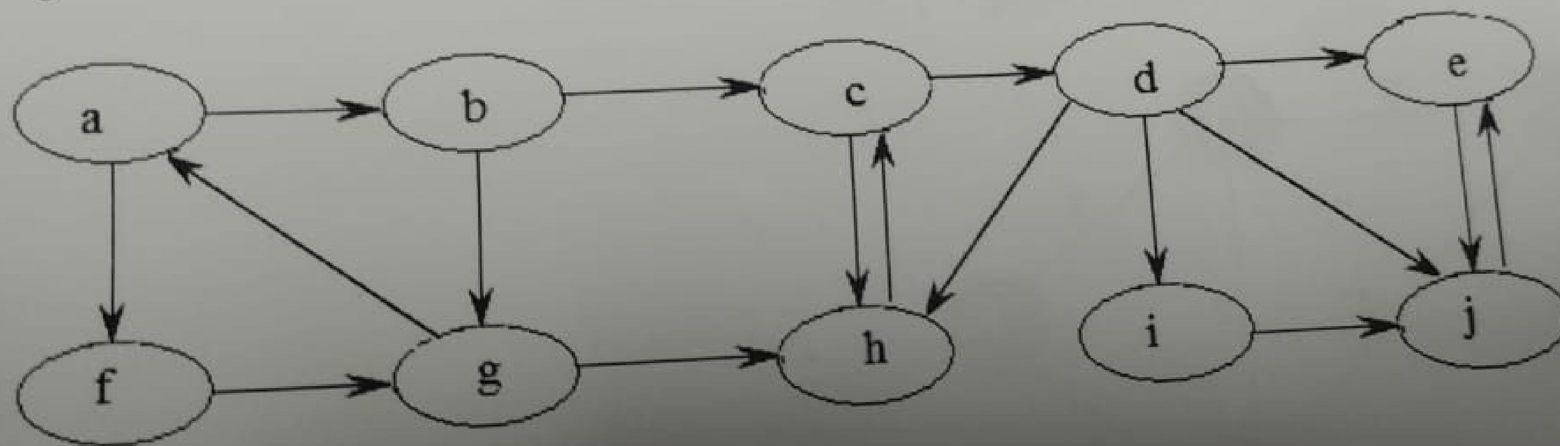
Find Single source shortest path from vertex ***a*** using Dijkstra.



Dijkstra:

Name: Roll: CT2-20 Marks

1) Identify different types of edges (forward edge, back edge, tree edge, cross edge, etc) using DFS graph search algorithm. Consider **a** as source. Also find SCC for the given graph.



Jahangirnagar University

Department of Computer Science and Engineering

2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2023

Course Title: Algorithms-I

Time: 3 Hours

Course No: CSE 209

Full Marks: 60

Read the question carefully.

There are five sections. Each section carries equal marks.

Figures in the right margin indicate marks.

1)

SECTION-1

Section No. 1 is based on *CO1*. There are six questions.

You have to answer **any six questions** from this section.

a) i) Define Algorithm. 2

ii) Differentiate between Big O, Big Omega and Big Theta notation in single sentences.

b) Show that $3n^5 + n^2 + 1 = \theta(n^5)$. 2

c) Recall the properties of a greedy algorithm. 2

d) Identify the limitations of binary search. 2

Can we perform binary search on the give sequence { 7, 3, 0, -1, -2}? After answering Yes or No, justify your answer.

e) Explain divide and conquer paradigm. 2

f) Differentiate between weakly connected component and strongly connected component. 2

g) Define prefix-free code with example. 2

2)

SECTION-2

Section No. 2 is based on *CO2*. There are four questions.

You have to answer **any three questions out of four** from this section.

a) Explain Merge sort algorithm with the following example. 4

{8 , 5, 7 , 2 , 9 , 1 , 6, 4, 3}

Also find the complexity of the algorithm using big O notation.

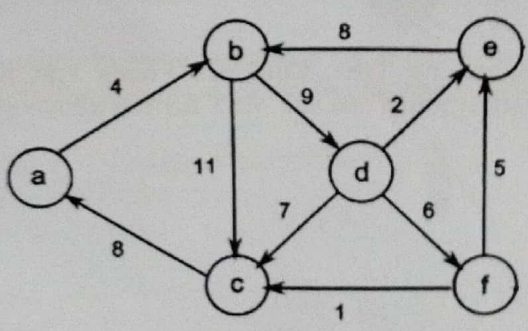
b) Explain the Quick sort with the following sequence as example (step by step). 4

{4, 6 , 2 , 8 , 1 , 5, 3}

Also find the complexity of the algorithm using big O notation.

c) Give an example of the integer knapsack problem (items with size and values, and a knapsack capacity) to show that the greedy fractional knapsack problem does not always work for solving the integer knapsack problem. 4

d) Determine the single-source shortest path for the following graph using Dijkstra's Algorithm, with the source node being with node a. 4



Also find the complexity of your solution using Big O notation.

SECTION-3

Section No. 3 is based on *CO3*. There are four questions.

You have to answer **any three questions out of four** from this section.

3) a) $f_n = (2f_{n-1} - 3\ f_{n-2} + 4\ f_{n-3})\ mod\ 59$ where $f_0 = 0,\ f_1 = 3,\ f_2 = 2.$ 2+2

i) Sketch the solution using matrix exponentiation for solving recurrence relation and show the base matrix.

ii) Calculate the result for n = 19 using matrix exponentiation.

b) Demonstrate step by step procedure of Heap Sort algorithm (using max heap) for the given sequence {5, 6, 3, 1, 4, 2}. 4

Also find the complexity of the algorithm using big O notation.

c) Given Text, X= "BACABACB" and Text, Y= "ACBAABAC". 4

Calculate the LCS between two text X and Y using tabular format. Also formulate a recursive solution and mention the complexity of the solution using big O notation.

d) A file has the following characters along with their frequencies: 4

d:31, e:21, c:18, a:15, m:12, p:5, q:2.

Draw Huffman coding tree and compute the optimal Huffman code of each character mentioned above.

SECTION-4

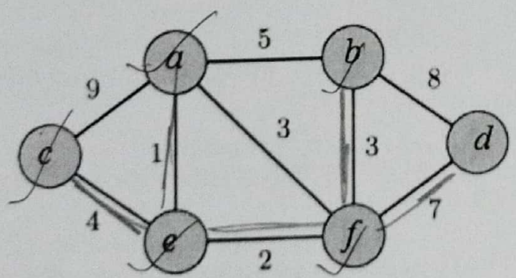
Section No. 4 is based on *CO4*. There are four questions.

You have to answer **any three questions out of four** from this section.

4) a) Compare the following graph traversal and pathfinding algorithms: BFS, DFS, Dijkstra, Floyd-Warshall, and Bellman-Ford Algorithm in terms of: 1× 4

i) Applications (e.g., single-source shortest path, reachability check, etc.)
ii) Complexity (asymptotic notation) in terms of time and space.
iii) Algorithmic Approach (e.g., greedy, dynamic programming, etc.)
iv) Handling of Graph Types (e.g., directed, undirected, weighted, unweighted)

b) 4



Define Minimum Spanning Tree. Compare Prims and Kruskal's algorithm for finding the MST. Hence, point out the MST for the above graph.

c) Compute the optimal price for the 0-1 knapsack problem for knapsack capacity, W=8. Which items will be in the optimal solution? What will be their total price? Show how you calculate the optimal solution using a matrix containing both price and arrows for each cell in the matrix. 4

item	A	B	C	D	E	F
Weight	1	2	3	2	4	3
price	7	8	14	5	10	15

d) A list of different activities with starting and ending times. Calculate the maximum size of non-overlapping activities that can be performed. 4

Activities	1	2	3	4	5	6	7	8
Start Time	1	0	1	4	2	5	3	4
Finish Time	3	4	2	6	9	8	5	5

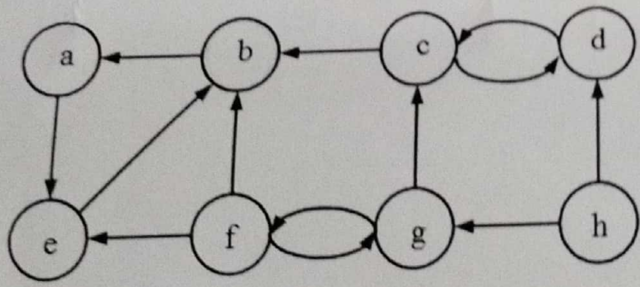
Also find the complexity of the solution using big O notation.

SECTION-5

Section No. 5 is based on *CO5*. There are three questions.

You have to answer **any two questions out of three** from this section.

5) a) Identify different types of edges (forward edge, back edge, tree edge, cross edge, etc) using DFS graph search algorithm. Consider **a** as starting vertex (node). Also find SCC for the given graph. 3+3



2× 3

b) $T(n) = T(n/4) + T(n/2) + n^2$

Formulate the theoretical complexity of the given recurrence relation using-

i) Substitution method

ii) Recursion tree method

iii) Master method

c) Estimate longest increasing subsequence from the given sequence using dynamic programming using tabular format. 3+3

$$S = \{10, 22, 9, 33, 21, 50, 41, 60, 80\}$$

Will this method be efficient for a sequence with for $N = 1{,}00{,}000$? If not, then formulate a better solution ($nlogn$) solution for this problem and demonstrate it.

---------o----------

Good Luck !!!!